
Self-Supervised Feature Ensemble for Robust Vision-Language-Action Models under Complex Backgrounds

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Abstract

Vision-Language-Action (VLA) models have shown great promise for generalist robot manipulation by leveraging pre-trained visual and language knowledge. However, their visual encoders often suffer from degraded attention in complex background textures, leading to poor task performance in cluttered or visually noisy environments. To address this problem without requiring manual annotations or model architecture changes, we propose a self-supervised feature ensemble method. Specifically, we extract DINOv2 features from multiple geometrically transformed copies of the same image, spatially shift them back, and average the results to obtain a stable, noise-robust feature target. Using 40,000 image-feature pairs sampled from the RoboTwin 2.0 benchmark (500 randomized episodes + 50 clean episodes), we fine-tune only the last 4 transformer blocks of the visual encoder while keeping the rest of the VLA model frozen. Experiments on RoboTwin 2.0’s randomized scenes show that our method significantly improves task success rate under complex backgrounds compared to the baseline OpenVLA-OFT, with minimal additional training cost. Our approach is annotation-free, architecture-agnostic, and computationally efficient, offering a practical solution for enhancing VLA visual robustness.

1 Introduction

Vision-Language-Action (VLA) models have emerged as a promising paradigm for generalist robot manipulation. By fine-tuning pre-trained vision-language models (VLMs) to directly output robot actions, models such as OpenVLA (1) and RT-2 (2) achieve impressive generalization across diverse tasks and embodiments, leveraging Internet-scale vision-language data and large-scale robot demonstration datasets. However, despite their strong performance in relatively clean environments, recent studies reveal that VLA models remain surprisingly vulnerable to visual distribution shifts (3). In particular, complex background textures, cluttered scenes, and lighting variations can cause the visual encoder — e.g., the DINOv2 component in OpenVLA — to attend to irrelevant background patterns rather than task-relevant objects, leading to significant drops in task success rate (1; 3).

Existing solutions either modify the architecture (dual encoders (4), alignment losses (5)), preprocess inputs (ARRO (6), CRT (7)), or regularize attention (gaze heatmaps (8), token pruning (9)). These often require extra models, manual labels, or architectural changes. Crucially, no prior work has explicitly analyzed how background textures corrupt DINOv2 attention inside VLAs, nor used such analysis to construct self-supervised training data.

In this paper, we address the following question: *Can we enable a VLA model to “learn to see more accurately” in complex backgrounds, using only its own outputs and without human supervision?* We propose a novel self-supervised feature distillation method built on a simple yet powerful observation: averaging features from multiple geometrically transformed views of the same image — after spatial

alignment — produces a more stable, noise-robust feature representation that suppresses background distractions. Concretely, we take a pre-trained OpenVLA model (1) and extract DINOv2 features from each input image under five geometric transformations: up translation, down translation, left translation, right translation, and left-right flip. After shifting the extracted features back to the original coordinate space, we average them to obtain a “stable target” feature. Using the RoboTwin 2.0 benchmark (13), which provides 12,000 high-quality background textures and domain-randomized scenes, we sample 40,000 image–feature pairs from 500 randomized episodes and 50 clean episodes. We then fine-tune only the last 4 transformer blocks of the DINOv2 visual encoder while freezing all other parameters, requiring only a single A100 GPU and 2,000 training steps. Our main contributions are threefold:

- We propose a new approach that enables the robot to learn to see more accurately on its own, which does not require manual annotation or modifications to the original model architecture.
- We construct a dataset containing 40k image–feature representation pairs derived from RoboTwin 2.0 (13), which can be reused for future research on visual robustness.
- With our proposed method, the original model gains performance improvements after a short training time, as validated on RoboTwin 2.0’s Hard (domain-randomized) scenes.

2 Related Work

Recent studies show VLA models are surprisingly vulnerable to visual distribution shifts (3). To improve robustness, strategies span model architecture, training, and input preprocessing. GenVLA (4) uses a dual vision encoder with frozen pre-trained representations. Kachaev et al. (5) apply an alignment loss to regularize visual features. ARRO (6) replaces backgrounds with virtual scenes; CRT (7) restores corrupted inputs via a plug-and-play Transformer; TraceVLA (10) overlays motion trajectories for spatio-temporal awareness.

Several works further investigate VLA robustness under specific visual disturbances and analyze attention mechanisms. A systematic benchmark (3) shows that camera view changes and initial robot states are the most severe factors causing performance drops, while background texture changes also significantly affect models like OpenVLA. To address attention diffusion in VLA models, Pani and Yang (8) propose gaze regularization training, using synthetic human gaze heatmaps with KL divergence to guide internal attention toward task-relevant regions. VLA-IAP (9) introduces interaction-alignment-based visual token pruning, retaining tokens critical for manipulation based on geometric and motion priors rather than semantic saliency. At the visual encoder level, research on ViT attention artifacts reveals that DINOv2 produces high-norm outlier tokens in low-information regions (e.g., uniform backgrounds), contaminating attention maps (11). Subsequent work (12) further elucidates the causal mechanism of these artifacts and proposes a training-free test-time register technique to eliminate them.

Despite these advances, existing methods either improve robustness in a general way from the architecture or training perspective, or target specific disturbances such as sensor noise, viewpoint shifts, or occlusions. No prior work has specifically investigated how complex background textures interfere with the attention patterns of DINOv2 within VLA models, nor has any work utilized attention-level analysis to construct more robust training data. To fill this gap, we propose to repeatedly feed scenes with complex background textures into the VLA model, extract the patch-level feature vectors from multiple geometrically transformed copies of each image, spatially align them back, and average the aligned features to obtain a stable, noise-robust feature target. These refined feature targets are then used to build an improved training dataset, and the model is fine-tuned to achieve robust manipulation performance in visually complex environments.

3 Methodology

We propose a self-supervised method to improve the visual robustness of OpenVLA in complex background textures. The pipeline consists of three stages: (1) generating stable feature targets via transformation ensemble, (2) constructing a dataset of image–feature pairs, and (3) fine-tuning only the last 4 blocks of the vision encoder with these targets.

3.1 Stable Feature Generation via Translation Ensemble

To suppress background-induced noise in the feature space, we leverage an ensemble of spatially translated copies of the same input image. For an input image x we create 5 geometrically transformed variants: up translation, down translation, left translation, right translation, and left-right flip. The translation amounts are fixed integer multiples of the patch size (14 pixels per patch), ensuring that the resulting feature maps can be precisely aligned back to the original patch grid. Each variant is passed through the frozen DINOv2 encoder to obtain its patch token features.

To align all features to the original coordinate frame of the input image, we apply the inverse spatial transformation to each extracted feature map, i.e., the “shift back and overlay” operation. This step ensures strict semantic consistency of the same visual region across all feature branches. Finally, we compute the element-wise average of all aligned feature maps to obtain the stable target feature:

$$F_{\text{avg}} = \frac{1}{N} \sum_{i=1}^N T_{-t_i}(F(x_{t_i})) \quad (1)$$

where N is the number of translated variants, T_{-t_i} is the inverse transformation for spatial alignment, and $F(x_{t_i})$ is the feature map extracted from the translated image. The resulting F_{avg} is a stable, noise-robust target that preserves task-relevant structures while averaging out background distractions.

3.2 Dataset Construction

We use RoboTwin 2.0 (13) as our data source. From its *randomized setting* (which includes diverse background textures, clutter, and lighting variations) we sample 500 episodes, and from the *clean setting* we sample 50 episodes. Applying the above pipeline to frames extracted from these episodes yields approximately 40,000 image–feature pairs (x, F_{avg}) . The entire data construction is fully automated and requires no human annotation.

3.3 Fine-Tuning the Last 4 Blocks of the Vision Encoder

Following the standard evaluation protocol of RoboTwin 2.0 (13), we first obtain a base model by fine-tuning OpenVLA (1) on 50 clean expert demonstrations using LoRA (denoted **OpenVLA-OFT**). On this base model, we freeze all parameters **except the last 4 transformer blocks of the DINOv2 encoder** (blocks 19–22) and its final layer norm — the earlier blocks of the visual backbone, the projection layer, the Llama 2 language model, and the action head all remain frozen.

The training objective is a combination of cosine similarity loss and mean squared error (MSE) loss, weighted by patch-level confidence. For each patch token, we compute:

$$\mathcal{L}_{\text{cos}} = 1 - \text{cosine}(F_p^{\text{pred}}, F_p^{\text{avg}}), \quad \mathcal{L}_{\text{mse}} = \frac{1}{C} \sum_{c=1}^C (F_{p,c}^{\text{pred}} - F_{p,c}^{\text{avg}})^2 \quad (2)$$

where $C = 1024$ is the feature dimension. The total loss for a sample is the sum of cosine and MSE terms, averaged over all valid patches, and each patch is weighted by its confidence factor $\text{valid_count}/5.0$. This design makes the model focus on high-confidence regions (typically the central areas that benefit from all ensemble branches) while down-weighting edge patches with fewer contributors.

3.4 Why Does This Method Work?

From a statistical perspective, extracting and averaging features from slightly translated images is essentially performing an ensemble in the feature space. By sampling within the input space, this strategy effectively neutralizes random noise, reduces feature variance, and provides a more stable and robust feature target.

By fine-tuning the last 4 blocks of the visual model with such features, the model can conduct task-specific adaptive optimization while retaining fundamental visual priors learned from pre-training. The frozen earlier blocks preserve general visual structures (edges, textures, local patterns), while the tuned later blocks adapt the attention to critical image regions relevant to the manipulation task. As a result, the fine-tuned vision encoder produces higher-quality feature representations that are both stable under domain randomization and compatible with the frozen action head of OpenVLA-OFT.

4 Experiments

We evaluate our method on the RoboTwin 2.0 benchmark (13), a bimanual robotic manipulation simulator providing two distinct settings: a **Clean Setting** with fixed backgrounds, no clutter, and standard lighting, and a **Randomized Setting** that applies full domain randomization across five dimensions (cluttered distractors, diverse background textures, varied lighting, table height adjustments, and diverse language instructions). Our training dataset consists of approximately 40,000 image-feature pairs generated automatically without manual annotation, sampled from 500 randomized episodes and 50 clean episodes.

Our baseline is **OpenVLA-OFT**, obtained by fine-tuning the official OpenVLA (1) on 50 clean expert demonstrations using LoRA for 100,000 steps on 8 A100 GPUs. We then apply our self-supervised method to this baseline. Following the exact protocol stated in our presentation, we freeze all components of OpenVLA-OFT except the last 4 Transformer blocks of the DINOv2 vision encoder. The model is trained on a single A100 GPU for 2,000 steps using the combined loss between the predicted features and the stable pseudo-labels generated via translation ensemble.

We compare our approach against five mainstream baselines evaluated under the same standard protocol. As shown in Table 1, even state-of-the-art models suffer significant degradation in randomized environments (e.g., DP drops to 0%, DP3 to 6%).

Table 1: Baseline success rates (%) on RoboTwin 2.0.

Model	Clean Setting	Randomized Setting
RDT	25%	12%
Pi0	58%	21%
ACT	22%	4%
DP	39%	0%
DP3	70%	6%

The experimental results demonstrate that our method consistently improves performance across most tasks. Table 2 summarizes the success rates under both the Clean and Randomized Settings, along with the corresponding relative improvements.

Table 2: Success rates and relative improvements (our method vs. OpenVLA-OFT baseline).

Task	Clean Setting (%)			Randomized Setting (%)		
	Original	Ours	Δ Clean	Original	Ours	Δ Randomized
1	41	44	+7.32%	16	21	+31.25%
2	41	40	-2.44%	17	20	+17.65%
3	36	41	+13.89%	14	18	+28.57%
4	34	39	+14.71%	18	19	+5.56%
5	44	47	+6.82%	15	17	+13.33%
Avg.	39.2	42.2	+7.65%	16.0	19.0	+18.75%

As shown in Table 2, our method achieves significant gains in the challenging Randomized Setting, with an average relative improvement of **18.75%**. In the Clean Setting, our model maintains comparable performance, with an average improvement of **7.65%** across the five tasks.

In conclusion, our method enhances the visual robustness of OpenVLA without modifying the model architecture or requiring manual labels. By fine-tuning only the last 4 blocks of the vision encoder for a short duration on a single GPU, we achieve significant gains in complex backgrounds while maintaining stability in clean scenes, proving the effectiveness of our self-supervised feature alignment strategy.

5 Discussion and Conclusion

Our experimental results demonstrate that the proposed self-supervised feature ensemble method effectively enhances the task success rate of OpenVLA-OFT. The observed performance gain is more pronounced in the challenging randomized setting (+18.75%) than in the clean setting (+7.65%). This pattern aligns with our underlying hypothesis: in environments with complex, distracting backgrounds, the DINOv2 encoder’s features exhibit higher variance, and our averaging process over translated views effectively mitigates this noise. In cleaner settings where the encoder is already relatively stable, the room for improvement is naturally smaller. While a marginal performance trade-off is noted in Task 2 (a slight decrease of 2.44% in the clean setting but a substantial 17.65% gain in the randomized setting), this is a reasonable compromise given the overall, significant boost in robustness against visual clutter.

A key advantage of our approach is its practicality. Unlike methods that introduce architectural modifications, require additional manual annotations, or incur extra computational overhead at inference time, our method fine-tunes only the last four blocks of a pre-trained visual encoder using self-supervised targets. The translation ensemble is used solely during target generation for training, leaving the inference pipeline unchanged and efficient. This makes our strategy a lightweight and broadly applicable plug-in for enhancing the visual robustness of existing VLA models.

5.1 Limitations and Future Work

While our method demonstrates promising results, several limitations point to avenues for future research. First, our intervention is confined to the visual encoder; the language understanding and action prediction components of the VLA remain frozen. Consequently, the overall performance gain is intrinsically bounded by the capacities of these fixed modules. Exploring lightweight, concurrent adaptation strategies for the entire model could yield further improvements. Second, our translation-based ensemble is a training-time technique. It does not directly address robustness to extreme, unseen viewpoints or heavy occlusions that fall outside the distribution captured by RoboTwin 2.0’s domain randomization. Extending the self-supervision paradigm to handle a wider range of geometric and semantic perturbations is an important next step.

5.2 Conclusion

In conclusion, we have introduced a simple, self-supervised feature ensemble method to improve the visual robustness of Vision-Language-Action models. By generating stable feature targets through the spatial alignment and averaging of features from multiple translated views, and fine-tuning only the last four transformer blocks of the visual encoder, we achieve a significant relative improvement (+18.75%) on domain-randomized benchmarks without altering the model architecture or inference process. Our work makes three concrete contributions: (1) a novel, annotation-free fine-tuning strategy for visual encoders, (2) a new dataset of 40,000 image-feature pairs constructed from RoboTwin 2.0, and (3) empirical validation demonstrating substantial gains with minimal computational overhead. This research underscores the potential of leveraging self-supervised, data-centric approaches to adapt powerful foundation models for more reliable and robust embodied AI.

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